

TRAINED AGENTIC CONTEXT MANAGEMENT

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ABSTRACT

We study long context language models. Instead of training long context natively, or designing a long context harness, we train a model over the simplest possible harness: a tool to call itself with any specified prompt and a tool to read tokens in a range from the input context. We finetune Qwen3.6-35B-A3B over this harness. With only 8,000 tokens of context, our small model outperforms GPT-5.4 with 1M tokens of context on the OOLONG-synth benchmark when document length exceeds 40K tokens.

1 INTRODUCTION

As language models boast impressive breakthroughs (Luong & Lockhart, 2025; Wei, 2025; OpenAI, 2025b; Lin & Cheng, 2025; OpenAI, 2026g;f;e), increasing focus is placed on their limitations. The most salient resource constraint of a language model is its context length. Models can process and produce tokens within this limit, but the limit is hard. Model producers have identified this constraint and placed great emphasis on extending it as large as possible, with some model builders touting 10M (Dangel, 2026; Meta AI, 2025; Zhu et al., 2026) and even 100M (Magic, 2024) token context lengths (for reference, the length of the first Harry Potter book is about 100,000 tokens, the full U.S. tax code roughly 3M, and the Linux kernel source code about 300M).

While native context scaling is appealing as a concrete engineering problem, it is not the only approach to producing a model capable of processing long documents. In 2021, OpenAI fine-tuned GPT-3 to summarize sections of books, and then summarize those summaries, achieving competitive performance with human book summaries on a model with a context length of just 2,048 tokens (Wu et al., 2021). As the field has matured, agentic harnesses have become viable. MemGPT treats context similarly to an operating system, relying on LLM-invoked function calls to push relevant information in and out of the context window to external memory, effectively achieving expanded virtual context (Packer et al., 2023). More recently, Recursive Language Models introduced a harness that utilizes LLM’s coding prior to call itself in a REPL environment, achieving stronger performance than base models on retrieval and aggregation tasks both within context limits, but also beyond it, expanding effective context by $\sim 4\times$ (Zhang et al., 2025).

While handcrafted prompts and programmatic environments can be effective in solving long context problems without native long context training, it begs an obvious question: if models can, for example, solve all the problems at the International Collegiate Programming Contest¹ World Finals (OpenAI, 2025b), or more generally, implement requested features in codebases far larger than context limits (Marten et al., 2026; OpenAI, 2026d; Anthropic, 2026b), can they not be *trained* to manage context effectively, *without human-authored harness crutches*? Such model training internalizes decisions external harnesses hardcode, offering the potential to scale much more generally.

We answer this question in this paper, affirmatively. We consider the simplest possible agentic context harness, defined below. The purpose of the harness is to *enable capability*, without *fixing choices* or *instilling a prior*.

1.1 HARNESS

Our harness has two tools:

¹<https://icpc.global/>

1. `read_chunk(start, end)`: Read a slice of the document and return the decoded text of tokens `[start, end)`.
2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child's `\boxed{}` answer.

Unmodified long-context prompts are provided to the model over a document accessed via the above harness. The system prompt is propagated to child agents and includes the above harness definition, a generic instruction, and potentially task-specific background, in accordance with the specified `eval2`. Subagents' user prompt is specified by the calling model.

1.2 RESULTS

We finetune Qwen3.6-35B-A3B on the above harness on a diverse synthetic data mixture and achieve the following results:

1. blah
2. blah blah

1.3 PAPER ORGANIZATION

In the next section, we survey related work to frame the contribution of the following sections. In Section 3, we describe our training methodology. In Section 4, we discuss all our experiments, and we conclude in Section 5.

2 RELATED WORK

Long-context LLM work can be broken into a few distinct categories: *native long context models*, *context compaction*, and *long context harnesses*, both trained and untrained.

2.1 NATIVE LONG CONTEXT MODELS

GPT-2 shipped with a 1,024 token context limit (Radford et al., 2019). This limit doubled in every generation through GPT-4 (Brown et al., 2020; OpenAI, 2022; 2023b), until jumping to the 100K range on GPT-4 Turbo (OpenAI, 2023a), with contemporaneous models Claude 2 (Anthropic, 2023) and later Llama 3.1 (Grattafiori et al., 2024) at similar limits. The core difficulty in scaling context length n in the classic Transformer architecture (Vaswani et al., 2017) is $\Omega(n^2)$ attention complexity, which becomes prohibitively expensive as $n \gg 100K$. State-space architectures like the mamba lineage (Gu & Dao, 2024; Dao & Gu, 2024; Lahoti et al., 2026), neural methods like Titans (Behrouz et al., 2024), and Linear Transformer (Katharopoulos et al., 2020) derivatives DeltaNet (Schlag et al., 2021) and Gated DeltaNet (Yang et al., 2025) abandon quadratic attention for recurrence. Modern architectures are hybrid, combining linear or sparse attention layers with dense attention, allowing context lengths to scale to 1M+ tokens while retaining long-context performance (Qwen Team, 2026; Kimi Team, 2026b; DeepSeek-AI, 2025; GLM-5 Team, 2026; NVIDIA, 2025; Gemma Team, 2024; Mistral AI, 2024).

2.2 CONTEXT COMPACTION

An alternative to native long context, and an inevitability when such native context fills, is a function f mapping current context x to a reduced context y where $|y| \ll |x|$. Three categories of such lossy f exist: *masking*, *summarization*, and *learned compaction*. The simplest compaction technique is masking: dropping tokens based on a predetermined rule. Clearing tool call results is a popular recommended choice (Anthropic, 2025a), achieving results competitive with summarization (DeepSeek-AI, 2025; Lindenbauer et al., 2025). Alternative approaches evict tokens based on attention statistics (Xiao et al., 2024; Zhang et al., 2023; Li et al., 2024). Summarization is the current production harness default, implemented in Claude Code, Codex, and Gemini CLI (Anthropic,

²Precise harness environment is given in Appendix A.

2025a; OpenAI, 2026a; Google, 2026; Barbaste et al., 2026). Trained compaction is a new frontier, described as part of GPT-5.1-Codex-Max (OpenAI, 2025a) and Cursor’s composer (Cassano & Rush, 2026) training. “Still” trains a model to compress KV-Cache directly (O’Neill et al., 2026).

2.3 LONG CONTEXT HARNESSES

As long as LLMs have proven useful, an evolving line of work has focused on how to expand their capabilities without touching model internals. The first efforts were known as prompt engineering (Phoenix & Taylor, 2024; Berryman & Ziegler, 2024), which, though necessary on pre-trained base models (Brown et al., 2020), became internalized through instruction-tuning and reasoning RL (Ouyang et al., 2022; Guo et al., 2025). Function-calling has lead to a new wave of such efforts, where domain-specific knowledge, tools, and multi-agent systems live on a scaffold that interacts with powerful LLMs known as a “harness” (Lopopolo, 2026; Barbaste et al., 2026).

Long-context harnesses originate with systems like MemGPT (Packer et al., 2023), RAPTOR (Sarathi et al., 2024), and Chain of Agents (Zhang et al., 2024), where the overall goal is to engineer systems that decompose documents into contexts independent LLM calls can read and aggregate. Recursive Language Models is a popular recent example that allows for single-level recursive LLM calls via a programmatic environment (Zhang et al., 2025). Follow-up work expands subagent capabilities (Lumer et al., 2026), trains shared parent/child policies (Kim & Ahmad, 2026), and incorporates self-reflection (Alizadeh et al., 2026).

Memagent (Yu et al., 2026) trains a harness that walks linearly through context, using RL and a harness prompt to determine what to carry from node-to-node. Similarly to our work, they also use a small (8K token) context while processing long (as much as 3.5M token) documents. However, they test primarily on the distribution they train, compare only against other small models, and hardcode execution path and prompts in the harness.

Frontier models now utilize multi-agent systems. Kimi K2.5 trains “agent swarm” (Kimi Team, 2026a): the model is trained to call subagents via reinforcement learning. Subagent trajectories are not trained. GPT-5.6 has also trained multi-agent behavior (OpenAI, 2026c), notably utilizing 10,000 agents to solve Navier-Stokes (OpenAI, 2026e). Anthropic, DeepMind, Meta, and xAI likely do similarly, with multi-agent systems touted in respective model and product releases (Anthropic, 2025b; 2026a; OpenAI, 2026b; xAI, 2026; 2025; Meta Superintelligence Labs, 2026; Google, 2025; Gottweis et al., 2026).

3 METHODS

We deliberately use the minimal tool definitions necessary to train agentic context management. However, our harness is sufficient to allow arbitrarily complex multi-agent decompositions. Decompositions in the literature fall into roughly three categories: *tree reduce* (Wu et al., 2021; Sarathi et al., 2024; Zhang et al., 2022), *sequential folds* (Yu et al., 2026; Zhang et al., 2024), and *agentic delegation* (Zhang et al., 2025; Kimi Team, 2026a; Anthropic, 2025b). We utilize all three strategies.

Without any training, models generally make poor use of our harness. Base Qwen3.6-35B-A3B, GPT-5.4, and Kimi K2.6 use subagent delegation sporadically, sometimes choosing to read the entire document in one context, even if context limit is specified to be less than document length, and other times delegating the exact original question to subagents. Opus 4.6 does slightly better, showing real context pressure awareness and document range subagent delegation. Unfortunately it delegates *too* much, blowing through \$300 of API credits building massive subagent trees on a 100-question eval, only to overcount³. Instead, we compare to base models supplied the entire document in context with native context limits, where performance is stronger.

In general, models have a *weak prior* for context decomposition. Pre-training is pure language continuation, and agentic post-training does not teach decomposition strategies. Where Recursive Language Models (Zhang et al., 2025) can rely on strong coding priors instilled in both pre- and post-training, decomposition on our harness needs to be taught.

³Opus 4.6 does appear to perform somewhat decently on our harness with unlimited context budget. Unfortunately, due to the massive subagent trees, running the full comparison without restriction is cost-prohibitive.

One idea might be to RL an open weight model on our harness. Unfortunately, exploration in LLM RL is far too weak to escape the weak decomposition prior at modest compute budgets. It is also unclear what reward to assign subagents, and if GRPO (Shao et al., 2024) or its derivatives (Yu et al., 2025; Liu et al., 2025; Zheng et al., 2025) are used, what group to compare subagent trajectories in.

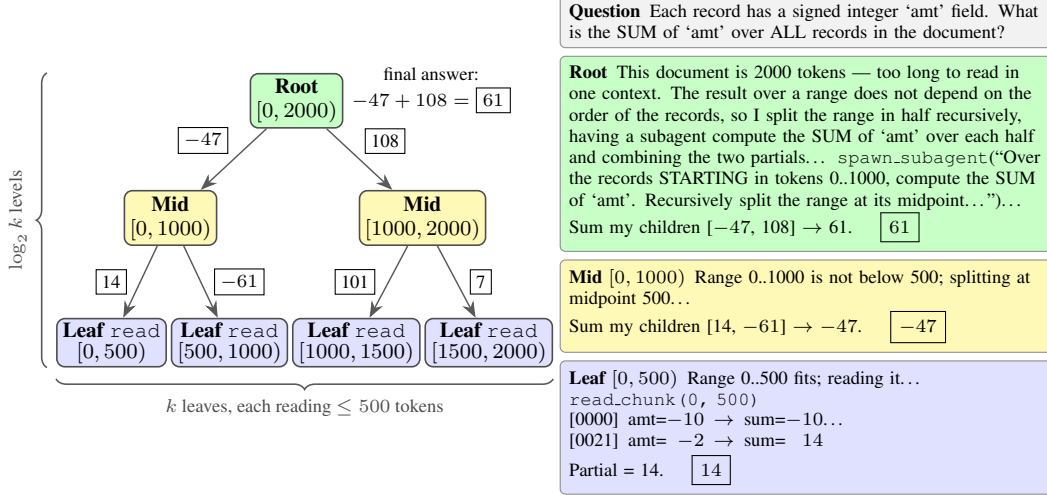


Figure 1: **Tree reduce**, for order-independent tasks (aggregation, retrieval). Non-leaf nodes delegate half their range to subagents. Leaves read one chunk and compute a leaf op which is aggregated up the tree.

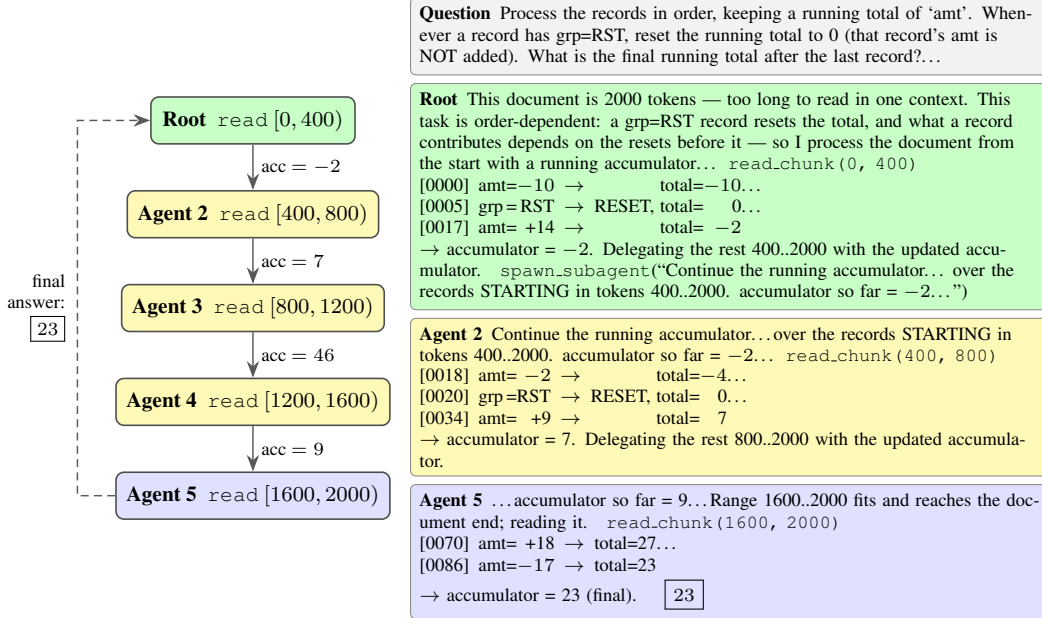


Figure 2: **Sequential fold**, for order-dependent tasks (variable tracking, running state). Each agent reads the next slice, updates the accumulator, and delegates the rest of the document to the next agent.

We instead SFT a decomposition strategy. We train *tree reduce* for order-independent tasks (see Figure 1), *sequential fold* for order-dependent tasks (see Figure 2), and *agentic delegation* to handle boundary conditions whenever we read the document (see Figure 3). By not hardcoding a decompo-

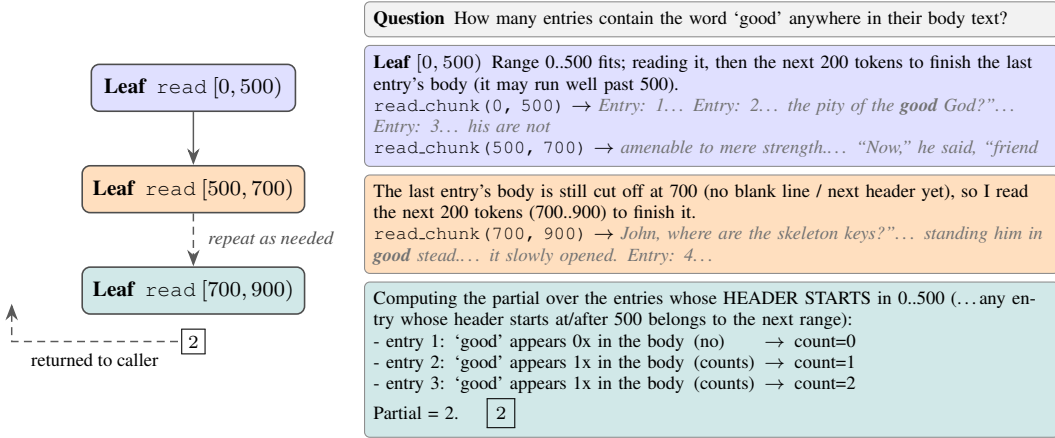


Figure 3: **Boundary handling.** A leaf/fold agent owns the data that *starts* in its range. It reads 200 additional tokens at a time until the last item in its range is not cut off (additional calls not pictured in Figures 1 and 2, for brevity).

sition structure in our harness, our model internalizes and composes delegation strategies according to the task and current problem-solving state.

We generate synthetic oracle demonstration data on a wide distribution of aggregation, retrieval, and sequential tasks. While we cover similar shapes to our evals, our training distribution uses independent questions and vastly more diverse examples. The technical challenge is to instill the necessary prior so the model knows how to take a question and delegate appropriate subtasks to agents from which the final answer can be determined.

For example, consider the training question “In how many sections are there *STRICTLY* more named than unnamed sentences?” (the prompt defines a *named* sentence as a sentence with a proper noun). To answer this question, each leaf must compute a tally of named and unnamed sentences per section in its assigned range. Thus, the root must recognize the necessary, often multi-dimensional, key space and dictate this and a recursive strategy to mid-level agents, who pass the instruction down to other mid-level agents, until leaves compute their range and counts are aggregated back up the tree, culminating in a final judgement at the root.

Our training data is primarily record-based, but also includes real prose, dot-separated word lists, and haystacks with injected needles. For more details on our complete training data, see Appendix B.

We do consider an RL stage following SFT. Unfortunately, even with LLM-as-a-judge rewards on subagent trajectories, as described in Recursive Agent Optimization Gandhi et al. (2026), the policy exhibits high-variance and latches onto simpler strategies that weaken overall performance. We suspect RL to be useful in learning recovery and strengthening generalization; however, to do this properly probably requires group-level baselines on subagent trajectories, to cut variance, and compute beyond the scope of this paper.

4 EXPERIMENTS

5 CONCLUSION

In this paper we show training on a bare bones agentic harness is sufficient for a small model with small context to outperform a frontier model with unlimited context on long-context evals. While we do not claim full generalization of our finetune, we also do not train on our evaluation distributions. The point of this paper is to show the power of agentic training in contrast to harness engineering, excessive native long context, and lossy compaction.

The thesis of the last decade of AI development is that specific model capability can be incorporated into fully-general models by training on a wider distribution. The methods of this paper can be

incorporated into frontier model development, so such models exhibit improved agentic performance under context pressure, without wasting tokens. This is in contrast to an external harness which must be applied at runtime.

To develop our curriculum into a fully-general product, the training distribution would need to be widened and a reinforcement learning stage would likely prove fruitful in supercharging agentic capabilities. We leave this as important future work, outside the scope of this paper or a small research project more generally.

AI USE STATEMENT

In this work, we used generative AI tools to write all our code, administer experiments, create figures, search and analyze relevant literature, format references, generate synthetic data sets, propose or refine hypotheses/brainstorming, design or provide feedback on research methodology/experiments, clean and reformat data, support qualitative and thematic data analysis, and interpret results. We have not used generative AI tools to draft parts of this research paper, and the following topics are not applicable to this work: help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, assist in the writing of proofs, and assist with translation. We have reviewed our AI-assisted work, for example, by reviewing AI code and conducting manual audits of synthetic data. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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A HARNESS AND PROMPTS

Benchmark	System prompt: task context	Root user message	Document
OOLONG-synth	OOLONG’s dataset description, verbatim (its intro, the number of items, the label set)	OOLONG’s question, verbatim, + “Put your final answer inside <code>\boxed{}</code> .”	One item per line: Date: ... User: ... Instance: ...
OOLONG-real	Everything before the first [START OF EPISODE], verbatim: the instructions and the player-to-character mapping	The dataset question, verbatim	The episode transcripts, verbatim
RULER	None (empty)	RULER’s task template, verbatim, with <code>{context}</code> replaced by a pointer to the document (below) + “Reply with the answer inside <code>\boxed{}</code> .”	RULER’s haystack with its inserted needles

Table 1: **Where each benchmark’s text goes.** The task context is placed in the system prompt, so every subagent inherits it without the parent restating it. For RULER, `{context}` in the template is replaced by: “[The relevant text is in a separate document accessible via the `read_chunk` tool — see the system prompt for usage.] (Document length: $\langle N \rangle$ tokens.)”

You are a long-document assistant. The document is $\langle N \rangle$ tokens long; you cannot see it directly. Your own context window is $\langle B \rangle$ tokens — the conversation (system prompt, user message, your responses, and all tool results) must fit in this budget or the episode ends.

$\langle$ task context, if the benchmark provides one $\rangle$

You have two tools:

- `'read_chunk(start, end)'`: read the document tokens in `[start, end)`.
- `'spawn_subagent(subtask)'`: delegate to a fresh-context copy of yourself (also $\langle B \rangle$ tokens) with `'subtask'` as the user prompt. The same tools and system prompt as above are inherited by the subagent.

When you are confident in the final answer, emit it as `\boxed{value}` and stop.

Tool descriptions (function-calling schema)

`read_chunk`: Read a slice of the document and return the decoded text of tokens `[start, end)`. The document has a fixed length (stated in the system prompt). Each call is capped at the chunk limit; for larger ranges, issue multiple reads or delegate to a subagent.

`start`: First token position to read (inclusive). `end`: Last token position to read (exclusive).

`spawn_subagent`: Spawn a fresh-context copy of yourself to solve `'subtask'`; returns the child's `\boxed{}` answer.

`subtask`: Sub-problem statement (free text) for the child to solve

Figure 4: **Agent system prompt**, shared by the root and every subagent, and the two tool descriptions. $\langle N \rangle$ is the document length and $\langle B \rangle$ the per-agent context budget. The root’s user message is the benchmark question (Table 1); a subagent’s user message is the subtask string its parent wrote.

System

$\langle$ task context, if the benchmark provides one $\rangle$

You are given a complete document followed by a question about it. Read the document carefully and answer the question. Show your reasoning, then emit your final answer as `\boxed{value}` and stop.

User

$\langle$ full document text $\rangle$

$\langle$ question $\rangle$

Figure 5: **Full-document prompt**, used for the single-shot baselines: the whole document is placed in one user message, with no tools and no budget prose. For RULER, whose template embeds the document mid-prompt, the document is instead placed back at the template’s `{context}` position, reproducing RULER’s own prompt.

B TRAINING CORPUS

You may include other additional sections here.